

Danny Thornewell

Seattle, WA | dthornewell@outlook.com | dthornewell.github.io/portfolio | (206) 503-3296

Education

Purdue University

August 2023 - May 2026

B.S. – Computer Science Honors, Minor in Mathematics

GPA: 3.95/4.0

Honors: Dean's List & Semester Honors (Fall 2023 - Spring 2025)

Relevant Coursework:

Completed: Linear Algebra, Discrete Math (Honors), Data Structures and Algorithms, Computer Architecture, Systems Programming, Analysis of Algorithms, Introduction to AI, Software Testing

In-Progress: Operating Systems, Computer Networks, Machine Learning, Robotics

Skills

Languages: Proficient in: **C++**, **C**, **Java**, **Python** – Familiar with: **C#**, TypeScript, SQL

Toolkits: CUDA, PyTorch, OpenCV, Flask, AWS SDK & CDK, SSL/TLS

Dev Tools: Git, GCC, GDB, Linux/Unix/Bash, Ghidra, Visual Studio, Docker, Postman

Cloud: AWS (ECS Fargate, Lambda, S3, CloudWatch, KMS, CloudFormation, API Gateway)

Experience

Amazon (Amazon Payments – Lumos)

May 2025 – Aug 2025

Software Development Engineer Intern

Seattle, WA

- Migrated a Tier-1 service handling Payment Critical information from Lambda to ECS Fargate.
- Increased API payload limits from 6 MB to 10 MB while reducing latency from 340 ms to 160 ms.
- Implemented mTLS authentication and met existing PCI-DSS & HIPAA security standards.
- Deployed to production traffic across 2 regions for 2 critical APIs using canary deployment strategies.

Sankar Research Lab

October 2024 – Present

Undergraduate Researcher

West Lafayette, IN

- Researching enzymatic retrosynthesis planners under Professor Karthik Sankaranarayanan.
- Developed Monte Carlo tree search (MCTS) based algorithm for synthesis pathway planning.
- Implemented MCTS with independent compound selection and parallelized simulation in Python
- Working with postdocs to utilize existing enzymatic reaction feasibility datasets for MCTS.
- Presented findings and progress with grant sponsors and worked with them to design reward functions aimed at reducing drug synthesis steps and starting chemical costs.

Autonomous Motorsports Purdue

January 2024 – Present

Software Team Lead

West Lafayette, IN

- Developed lightweight lane detection algorithms in OpenCV optimized for low-power microcontrollers.
- Won 2nd place for engineering posters at the Purdue Spring Undergraduate Research Symposium.
- Trained and finetuned segmentation computer vision models using PyTorch and ONNX to detect roads.
- Engineered an algorithm to curve-fit the track edges and generate waypoints in real-time using CUDA.
- Created new onboarding documents and stack documentation to help new members joining the club.

Projects

Reversi AI Engine

- Developed a fully playable Reversi (Othello) game in C++ with a CLI and GUI using the SDL2 library.
- Implemented several AI strategies for the computer opponent including MCTS, Minimax, & Negascout.
- Designed and trained a neural network in PyTorch as the board evaluation function to guide searches.
- Applied multithreading to parallelize MCTS simulations and Minimax search, improving search depth.